

Recovery Trajectory of Range of Motion and Its Association With Patient Factors During Early Postoperative Stiffness After Arthroscopic Rotator Cuff Repair

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Background: Postoperative shoulder stiffness may influence functional recovery after arthroscopic rotator cuff repair. However, predictors for progression to persistent shoulder stiffness have not been clearly established.

Purpose: To identify factors associated with persistent stiffness and describe range of motion (ROM) recovery trajectory after arthroscopic repair of full-thickness rotator cuff tears.

Study Design: Cohort study; Level of evidence, 3.

Methods: We retrospectively reviewed 352 patients who underwent arthroscopic complete repair. Stiffness was defined as forward flexion <120°, external rotation <30°, or internal rotation below L3. ROM, visual analog scale (VAS) score, American Shoulder and Elbow Surgeons score, Constant score, and tendon integrity were assessed, and patients reported the primary barrier to rehabilitation at 3 months. Predictors of persistent stiffness at 6 months were evaluated using multivariate logistic regression.

Results: At 3 months, 165 patients (46.9%) met stiffness criteria. Compared with patients without stiffness, the stiffness group had poorer functional outcome ($P < .001$), higher pain (VAS, 2.2 ± 1.1 vs 1.8 ± 1.2 ; $P = .002$), and a lower retear rate (6.7% vs 14.4%; $P = .019$). Among patients with early stiffness, 34 (20.6%) had persistent stiffness at 6 months. Fear or anxiety about tendon retear was the most common barrier (50.3%), followed by pain (32.7%); however, persistent stiffness was most prevalent when pain was the primary barrier (67.6%). Higher pain score at 3 months independently predicted persistent stiffness (odds ratio, 1.44 [95% CI, 1.01-2.05]; $P = .044$).

Conclusion: Early stiffness after arthroscopic rotator cuff repair is heterogeneous. Pain-driven stiffness is associated with persistence, whereas stiffness related to fear or anxiety about tendon retear is common but largely reversible; tailored management may improve recovery.

Keywords: arthroscopic rotator cuff repair; pain; postoperative shoulder stiffness; range of motion

Postoperative shoulder stiffness is one of the most common complications after arthroscopic rotator cuff repair, and it occurs in 11% to 35.4% of patients.⁵ It may lead to distressing and unsatisfactory situations for both patients and surgeons during the recovery period, despite successful rotator cuff repair.^{6,11,18,33} However, stiffness during the early postoperative period is associated with improved tendon healing and a lower retear rate.^{12,13,23,27,33} Patients with postoperative stiffness have functional deficits during the early recovery period; however, these typically improve

over time, and their long-term outcomes are generally comparable to those of patients without stiffness.^{3,17,25,27}

Despite the favorable long-term outcomes of early postoperative stiffness, the recovery trajectory of the shoulder range of motion (ROM) varies among individuals. Approximately 20% of patients with early postoperative stiffness have persistent stiffness beyond 6 months, indicating interindividual variability in recovery patterns.^{6,16} Several studies have suggested greater postoperative pain in patients with persistent stiffness after arthroscopic rotator cuff repair.^{11,25,34} However, most have used a cross-sectional approach; they determined the relationship between pain and postoperative stiffness based on ROM measurements at a single time point and compared pain levels longitudinally. This approach limits the evaluation

of pain progression relative to ROM recovery. Pain sensitization and kinesiophobia have recently been proposed as contributing mechanisms underlying the persistence of stiffness and poorer clinical outcomes.¹⁴ The temporal pattern of postoperative pain may influence ROM recovery; however, its role in determining patients who will progress to persistent stiffness has not been clearly established. The factors that become predictive of persistent stiffness at specific time points also remain unknown.

This study aimed to identify the factors affecting the persistence of early postoperative stiffness and the trajectory of ROM recovery in patients with full-thickness rotator cuff tears after arthroscopic repair. We hypothesized that greater early postoperative pain would be associated with persistent stiffness and slower ROM recovery.

METHODS

Patient Selection

The patients who underwent arthroscopic rotator cuff repair for symptomatic full-thickness rotator cuff tears at one institution between February 2019 and March 2023 were retrospectively analyzed. The inclusion criteria were as follows: having undergone arthroscopic rotator cuff repairs with full coverage of footprint for symptomatic full-thickness rotator cuff tears detected on preoperative magnetic resonance imaging (MRI), completion of follow-up evaluations at 1, 3, 6, 12, and 24 months, and confirmation of the integrity of the repaired tendon using MRI and ultrasonography performed at 6 months and 1 year postoperatively to confirm the repaired tendon integrity. The exclusion criteria were as follows: patients with preoperative shoulder stiffness, primary arthroscopic repair for partial-thickness rotator cuff tear or isolated subscapularis tear; additional augmentation procedures, such as patch augmentation or superior capsular reconstruction; revision surgeries; advanced glenohumeral osteoarthritis confirmed by imaging studies; neurologic or systemic disorders affecting the shoulder joint; history of acute trauma or infection; and incomplete medical records.

Postoperative stiffness during the recovery period was defined based on the presence of any of the following criteria at the follow-up evaluation: forward flexion of $<120^\circ$, external rotation of $<30^\circ$, or internal rotation below the L3 level.^{26,33} The patients were categorized into those

with (3-month stiff group) and without (3-month nonstiff group) stiffness based on their ROMs at 3 months postoperatively. The patients in the 3-month stiffness group were further categorized into those with stiffness (6-month stiff group) and those who had recovered from it (6-month nonstiff group) based on their status at 6 months postoperatively using the same criteria. This retrospective study was approved by the institutional review board of Ewha Womans University Seoul Hospital.

Clinical Evaluations

All patients completed the questionnaires before surgery. The questionnaires included questions about arm dominance, duration of shoulder symptoms, and the presence of diabetes mellitus, hyperlipidemia, and smoking. Body mass index (BMI) was calculated by dividing the body weight in kilograms by the square of the height in meters. The ROM in 3 directions, American Shoulder and Elbow Surgeons (ASES) score, Constant score, and pain severity based on the visual analog scale (VAS) score were evaluated preoperatively, at 3, 6, 12, and 24 months postoperatively, and at the final follow-up in the outpatient clinic by a single experienced physician assistant. The shoulder ROM was determined based on the passive ROM in 3 planes: forward flexion, external rotation at the side, and internal rotation behind the back. The measurements were performed using a goniometer by a single experienced physician assistant. Forward flexion was defined as the angle formed between the arm and the thorax with the elbow extended. External rotation at the side was defined as the angle between the thorax and the forearm when the upper arm was adducted, and the elbow was flexed at 90° . Internal rotation behind the back was defined based on the vertebral level reached by the tip of the thumb.⁶ The vertebral levels were converted into a contiguous numerical scale (higher values indicate better internal rotation) for ease of statistical analysis:²⁶ T1 to T12, 19 to 8; L1 to L5, 7 to 3; sacrum, 2; buttock, 1; and thigh, 0.

The patients with shoulder stiffness at the 3-month postoperative follow-up were interviewed to determine the reasons for poor compliance with rehabilitation exercises. Their responses were as follows: (A) pain; (B) lack of time or motivation; and (C) fear or anxiety regarding tendon retear. The patients were stratified into the corresponding subgroups, and the proportions of those with persistent stiffness at 6 months postoperatively were determined.

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Ethical approval for this study was obtained from Ewha Womans University Seoul Hospital Institutional Review Board (IRB No. SEUMC 2025-08-004). The data that support the findings of this study are not publicly available due to privacy and ethical restrictions but are available from the corresponding author upon reasonable request and with the approval of the institutional review board of Ewha Womans University Seoul Hospital.

Radiologic Evaluations

All patients underwent postoperative MRI at 6 months and ultrasound examination at 1 year to assess tendon integrity, regardless of symptoms. Tendon integrity was evaluated using the Sugaya classification, and retear was defined as type 4 or 5, indicating complete discontinuity of the repaired rotator cuff tendon.^{31,33} The pre- and postoperative MRI scans were interpreted by a musculoskeletal radiologist who was not involved in this study.

Intraoperative Evaluations

The tear size and pattern were determined intraoperatively after debridement of the degenerative tissue at the torn edge of the rotator cuff using a calibrated measuring device. The anteroposterior (AP) tear size was defined as the distance along the lateral edge of the exposed footprint. The mediolateral (ML) tear size was defined as the longest distance between the most medial margin of the retracted tendon and the footprint. The tear patterns were classified as crescent-shaped, U-shaped, anterior L-shaped, and posterior L-shaped. A crescent-shaped tear was defined as having an AP tear size larger than the ML tear size. A U-shaped tear was defined as having an ML tear size larger than the AP tear size. An anterior L-shaped tear was defined as a tear that propagated through the rotator interval and anterior rotator cable, resulting in exposure of the long head of the biceps tendon. A posterior L-shaped tear was defined as a tear that propagated along the interval between the supraspinatus and infraspinatus tendons.

Rehabilitation

All patients were immobilized with an abduction brace for 1 month postoperatively, regardless of tear size or retraction. They were allowed to perform elbow, wrist, and finger exercises immediately after surgery. No shoulder-specific physical therapy was administered during the immobilization period. Passive ROM exercises were initiated at 1 month postoperatively, followed by gradual active-assisted motion based on pain tolerance. Muscle-strengthening exercises were initiated for the patients who had achieved >80% of the ROM of the contralateral shoulder at 3 months postoperatively. Rehabilitation was performed with home exercises, which were instructed by a single experienced physician assistant, and patients performed the exercises independently at home.

Statistical Analysis

The categorical variables (sex, operated side, tear type, history of diabetes or hyperlipidemia, smoking status, and retear) were compared using the chi-square test. The continuous variables (age, symptom duration, BMI, tear size, VAS score, and ASES score) were compared using Student *t* test. Forward flexion, external rotation, and internal rotation were measured at 3, 6, and 12 months

postoperatively to assess the recovery trajectories of shoulder ROM in the stiff and nonstiff groups. Their recovery trajectories were compared using repeated-measures analysis of variance. Greenhouse-Geisser corrections were applied when the assumptions of sphericity were violated. The Mann-Whitney *U* test with Bonferroni correction was used for the intergroup comparisons at each time point because variance homogeneity was not met at several postoperative time points. Variables with $P < 0.1$ in the univariate analysis were entered into the multivariate logistic regression model. A post hoc power analysis was performed to assess the adequacy of the current sample size for the multivariable logistic regression model, based on the observed sample size, number of outcome events, and the effect size of the primary predictor of interest. Statistical significance was set at $P < .05$. All statistical analyses were performed using SPSS software, Version 29.0 (IBM Corp).

RESULTS

Of the 643 patients, 291 were excluded because of preoperative stiffness ($n = 153$), isolated subscapularis tendon tears ($n = 13$), and revision surgeries ($n = 19$). The 106 irreparable cuff tears that required additional graft augmentation procedures were also excluded. A total of 352 patients were included in the study. Of these, 165 (46.9%) and 187 (53.1%) were classified into the 3-month stiff and nonstiff groups, respectively. The demographic variables of the 2 groups were comparable, except for a higher proportion of female patients in the stiff group (Table 1).

The 3-month stiff group had significantly limited ROM in all 3 planes, lower ASES and Constant scores, and higher VAS scores than the nonstiff group ($P < .001$) (Table 2). The retear rate was significantly lower for the patients in the 3-month stiff group than for those in the 3-month nonstiff group ($P = .019$).

Of the 165 patients with early postoperative stiffness, 34 (20.6%) had persistent stiffness at 6 months, whereas 131 (79.4%) recovered full ROM. The 6-month stiff group had significantly more limited ROM and lower ASES (81.5 ± 8.6 vs 89.3 ± 7.3 ; $P < .001$) and Constant (71.5 ± 9.4 vs 83.8 ± 7.4 ; $P < .001$) scores at 6 months postoperatively and higher VAS scores at 3 months postoperatively (2.4 ± 1.3 vs 2 ± 1.1 ; $P = .040$) than the 6-month nonstiff group. However, their VAS scores did not differ significantly at 6 months (0.79 ± 0.88 vs 0.65 ± 0.83 ; $P = .371$).

Both 3-month stiff and nonstiff groups showed significant improvements in ROM over time ($P < .001$); however, their recovery trajectories differed significantly (forward flexion and external rotation: $P < .001$; internal rotation: $P = .011$) at 12 months. Consequently, no significant differences in ROM were observed between the groups at 24 months (Figure 1).

The interviews conducted at 3 months postoperatively revealed that fear or anxiety about tendon retear was the most frequently reported cause of poor adherence to

TABLE 1
Comparison of the Demographic Data of the 3-Month Stiff and Nonstiff Groups^a

Variables	3-Month Stiff Group	3-Month Nonstiff Group	P
No. of patients	165 (46.9)	187 (53.1)	
Age, years	62.4 ± 8.2	61.9 ± 9.3	.645
Sex, male:female	58:107	92:95	.008
Operated side, right/left	122/43	127/60	.215
Symptom duration, mo	23 ± 30	22.7 ± 41.1	.944
Diabetes mellitus	23 (13.9)	20 (10.7)	.354
Dyslipidemia	72 (43.6)	72 (38.5)	.328
Smoking	17 (10.3)	14 (7.5)	.352
BMI, kg/m ²	25.1 ± 3	25.5 ± 3.1	.217
Tear size, mm			
AP	22.3 ± 5.7	22.4 ± 5.5	.885
ML	22.8 ± 6.6	23.7 ± 6.8	.228
Tear pattern			.529
Crescent	49 (29.7)	58 (31)	
U	35 (21.2)	50 (26.7)	
L	28 (17)	25 (13.4)	
Reverse-L	53 (32.1)	54 (28.9)	

^aData are presented as mean ± SD or n (%). AP, anteroposterior; BMI, body mass index; ML, mediolateral.

TABLE 2
Comparison of the Clinical Outcomes of the 3-Month Stiff and Nonstiff Groups at 3 Months Postoperatively^a

Variables	3-Month Stiff Group	3-Month Nonstiff Group	P
ROM			
Forward flexion, deg	115.6 ± 24.5	147.4 ± 17.8	<.001
External rotation, deg	36.2 ± 19.8	56.2 ± 17.5	<.001
Internal rotation	4.1 ± 1.9	6.3 ± 1.7	<.001
ASES score	70.3 ± 9.8	75.9 ± 9.8	<.001
Constant score	81.2 ± 9.4	84.2 ± 8.7	.001
VAS score	2.2 ± 1.1	1.8 ± 1.2	.002
Retear, %	11/165 (6.7)	27/187 (14.4)	.019

^aData are presented as mean ± SD or n (%). ASES, American Shoulder and Elbow Surgeons score; ROM, range of motion; VAS, visual analog scale.

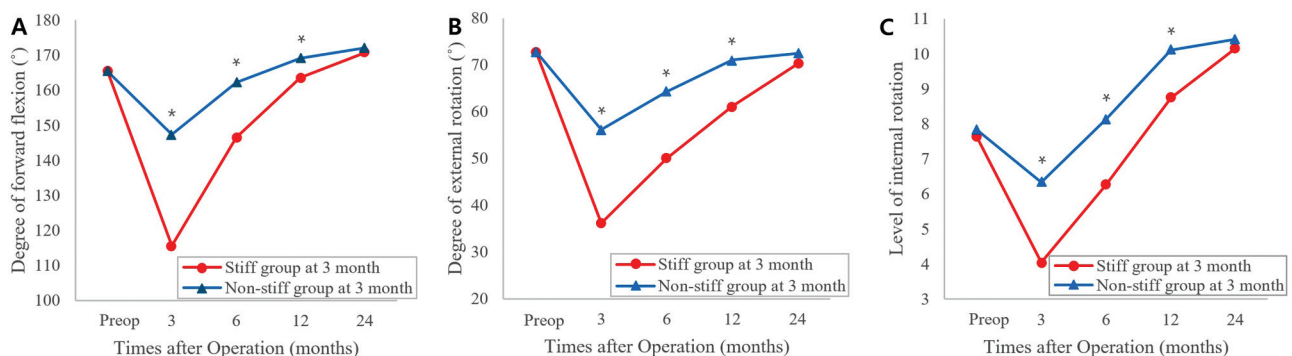


Figure 1. Comparison of the trajectories of ROM recovery for the 3-month stiff and nonstiff groups. (A) Forward flexion. (B) External rotation. (C) Internal rotation. An asterisk indicates a statistically significant difference between groups in ROM ($P < .05$). ROM, range of motion.

TABLE 3
Distribution of Reported Barriers to Rehabilitation Among Patients With Shoulder Stiffness
at 3 and 6 Months Postoperatively^a

Reasons for Poor Compliance to Postoperative Rehabilitation at 3 Months	Stiffness at 3 Months Postop n = 165	Persisted Stiffness at 6 Months Postop n = 34
Pain, %	54 (32.7)	23 (67.6)
Lack of time or motivation, %	28 (17)	5 (14.7)
Fear or anxiety about tendon retear, %	83 (50.3)	6 (17.6)

^aData are presented as n (%). Postop, postoperatively.

rehabilitation exercises, followed by pain (Table 3). However, the proportion of patients with persistent stiffness at 6 months postoperatively was highest for those who cited pain as the primary reason. In contrast, those who reported fear or anxiety had the lowest rate of persistent stiffness. Of the 54 patients who cited pain as the primary barrier at 3 months, 23 patients (42.6%) had persistent stiffness at 6 months. This proportion was substantially lower for those who reported a lack of motivation (17.9%) or fear or anxiety regarding tendon retear (7.2%). Eight patients who reported severe pain underwent injections of ropivacaine combined with corticosteroids into the intra-articular joint, subacromial space, or biceps tendon sheath.

The percentages represent the distributions of the reported barriers within each stiffness group at the corresponding time points and do not indicate the risk of persistent stiffness.

The operated side ($P = .070$) and the VAS score at 3 months postoperatively ($P = .040$) were included in the multivariate logistic regression analysis to identify the factors associated with persistent stiffness at 6 months postoperatively. The VAS score at 3 months postoperatively was the only independent factor associated with persistent stiffness at 6 months postoperatively ($P = .044$; odds ratio, 1.44 [95% CI, 1.01-2.05]). The post hoc power analysis indicated that the minimum detectable odds ratio at 80% power was 1.56 for the current sample. In contrast, the adjusted odds ratio observed for the VAS score at 3 months postoperatively was 1.44, which did not reach this threshold.

DISCUSSION

The most important finding of this study was that fear or anxiety regarding tendon retear was the most common reason for poor rehabilitation adherence at 3 months postoperatively. However, more severe pain was the primary factor associated with persistent stiffness at 6 months. These findings suggest that early postoperative stiffness is a heterogeneous condition driven by distinct patient-related factors rather than a uniform pathological entity. Therefore, individualized postoperative management strategies are required.

Postoperative shoulder stiffness after rotator cuff repair has been widely reported, with its incidence varying depending on the definition and timing of assessment.⁵ It

is most commonly observed during the early postoperative period and tends to decrease over time with continued rehabilitation.^{3,17,25,27} Although early stiffness may be associated with delayed recovery, several studies have suggested that it may also be associated with improved tendon integrity and lower retear rates, highlighting its complex clinical implications.^{12,13,23,27,33}

Pain was a significant patient-related factor associated with the persistence of postoperative stiffness in patients with arthroscopic rotator cuff repair.^{28,30} The relationship between postoperative pain and persistent stiffness can be better understood in the context of basic science. Inflammatory cytokines and neurogenic mediators such as substance P contribute to fibrotic pathways and restricted motion by promoting nociceptive sensitization and transforming growth factor β -mediated fibrosis.^{8,11,29} Namdari and Green²⁵ reported a significant correlation between pain severity and limited shoulder motion at 3 months postoperatively. However, their analysis was cross-sectional and limited to a single postoperative time point. Therefore, it did not evaluate the persistence or longitudinal progression of postoperative stiffness. The present study was distinguished by its longitudinal analysis of patients with early postoperative stiffness. It specifically evaluated the temporal evolution of the condition from 3 to 6 months to identify factors associated with its persistence. Pain was the most frequently reported barrier to rehabilitation at 3 months, accounting for 67.6% of the patients who had stiffness at 6 months postoperatively. Furthermore, 42.6% of the patients who cited pain as the primary barrier at 3 months had persistent stiffness at 6 months, which was substantially higher than the proportions of those who reported lack of motivation or fear or anxiety regarding tendon retear. Active pain control and anti-inflammatory treatment, such as nonsteroidal anti-inflammatory drugs or appropriately timed intra-articular corticosteroid injections, may help alleviate pain and facilitate the resolution of stiffness in patients with early postoperative stiffness accompanied by severe pain.^{1,9,20}

However, pain was not the most common reason for poor rehabilitation compliance at 3 months postoperatively in our cohort. The most common cause of low compliance with the postoperative home rehabilitation protocol at 3 months postoperatively was anxiety related to the possibility of a retear. Fear of movement and avoidance behavior can substantially limit active participation in rehabilitation and delay the recovery of shoulder mobility even after

successful tendon healing. Strong associations of low self-efficacy, low treatment expectations, and anxiety with reduced engagement in postoperative exercises have also been reported.⁴ Qualitative research has further supported this, showing that psychological and motivational factors strongly influence postoperative exercise compliance.³⁰ Many patients have reported fear of damaging the repair as an important barrier to active exercises. Anxiety about the possibility of a retear was the most frequently reported barrier to rehabilitation at 3 months postoperatively in the present study. However, only 6 of the 83 patients (7.2%) had persistent stiffness at 6 months. The low persistence rate for the 3 reported barriers may be explained by the reassurance provided during routine postoperative follow-up, which included information on the association of early postoperative stiffness with a lower retear rate and encouragement to continue rehabilitation. This finding highlights the potential reversibility of anxiety-driven stiffness with appropriate education and reassurance of possible low retear rates. It distinguishes it from pain-driven stiffness, which appears to reflect a different underlying mechanism.

Several studies have examined the relationship between early postoperative shoulder stiffness and retear rate after rotator cuff repair.^{23,27,33} Although the definitions of stiffness and the timing of retear assessment varied slightly among the studies, the literature consistently shows that patients with postoperative shoulder stiffness have better tendon integrity and lower retear rates than those without stiffness. This is explained by excessive loading or tension exceeding the mechanical strength of the repaired tendon before immature and disorganized type 3 collagen fibers are replaced by mature, well-aligned type 1 collagen along the direction of tensile stress.^{10,24} Parsons et al²⁷ reported that patients with stiffness during the early postoperative period had lower retear rates but similar ROM and functional scores at 1 year relative to those without stiffness. However, the observed difference in retear rates was not statistically significant. The study was limited by its small sample size and the absence of an analysis of factors contributing to stiffness. Our larger cohort not only confirmed the association between early stiffness and tendon integrity but also demonstrated complete resolution of ROM differences by 24 months. It also identified pain severity, kinesiophobia, and retear-related anxiety as key factors influencing the persistence of stiffness and rehabilitation compliance. In addition, previous studies have suggested that systemic inflammatory markers, such as the lymphocyte-to-monocyte ratio, may be associated with postoperative outcomes after rotator cuff repair.³² However, these studies primarily evaluated overall recovery at final follow-up using composite outcome measures and defined stiffness based on limited parameters. Therefore, their findings may not be directly applicable to the persistence of early postoperative stiffness evaluated in the present study.

Postoperative rehabilitation in this study was conducted using a home-based exercise protocol. Previous studies have demonstrated no significant differences in pain severity, clinical outcomes, or retear rates between home-based exercise and institution-based supervised

rehabilitation after rotator cuff repair.^{2,15,21} Home rehabilitation may be a cost-effective and accessible strategy that facilitates patient participation. However, a major limitation of home rehabilitation is its reliance on patient self-regulation. The accuracy of exercise performance and progression of exercise intensity depend on the patient, and there is limited feedback to address fear or lack of motivation. Burns et al⁴ reported a significant decrease in adherence to home rehabilitation over time based on objective quantification using smartwatch-based inertial sensors. However, improvements in pain and clinical outcomes were associated with the degree of physical therapy participation. Various motivational strategies, including motivational interventions and motivational interviewing, have been introduced to enhance patient participation in rehabilitation programs.²² However, systematic reviews have demonstrated that these approaches did not result in statistically significant improvements in rehabilitation attendance, despite trends toward improved adherence ($P = .07$).^{22,35} More recently, digital interventions incorporating mobile applications, text messaging, or web-based platforms have been explored to support home exercise programs, with several studies reporting improved short-term adherence, although long-term effects remain uncertain.¹⁹ Building on this concept, digitally assisted rehabilitation programs have been proposed as adjunctive tools to provide objective monitoring and feedback, with early studies suggesting favorable outcomes in selected functional parameters.⁷ Of the patients in the early postoperative stiffness group who had poor rehabilitation adherence, 17% of patients reported lack of time or motivation as the primary barriers to exercise participation. These findings provide clinical context for recent reports on adjunctive systems designed to support home-based rehabilitation. They may help overcome psychological barriers to exercise participation in selected patients.

This study has some limitations. First, the data were collected prospectively; however, the study design was retrospective. Second, the potential influence of additional procedures, such as biceps tenotomy or distal clavicle resection, could not be fully controlled. Previous studies have reported that these procedures do not substantially affect postoperative shoulder stiffness.^{6,33} However, some degree of confounding cannot be entirely excluded. Third, adherence to home rehabilitation was assessed only through subjective patient self-reporting, and no objective monitoring tool was available for the quantitative evaluation of exercise performance. Fourth, errors of measurement of the ROM are inevitable. Subtle variations in arm positioning may introduce inaccuracies despite all measurements being performed by a single experienced assistant to minimize interobserver variability. Fifth, the present study may not have achieved adequate statistical power to stably detect an odds ratio of the observed magnitude for the multivariable logistic regression analysis, as reflected by the marginal P value and wide confidence interval of the primary finding. Larger prospective studies with greater sample sizes are warranted to more precisely estimate the effect size and confirm the reproducibility of these findings.

CONCLUSION

Fear or anxiety of tendon retear at 3 months and pain as the primary reason for persistent stiffness at 6 months were associated with poor home exercise compliance. Differentiating pain-driven stiffness from fear- or motivation-related avoidance and implementing tailored postoperative management strategies may help improve functional outcomes.

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